

1 Summary

Model (Run-name): model1 (test)

Iteration: 2

TAC: 128009.91260607763

Took: 0.13115978240966797s

Total Time: 0.4580564498901367s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

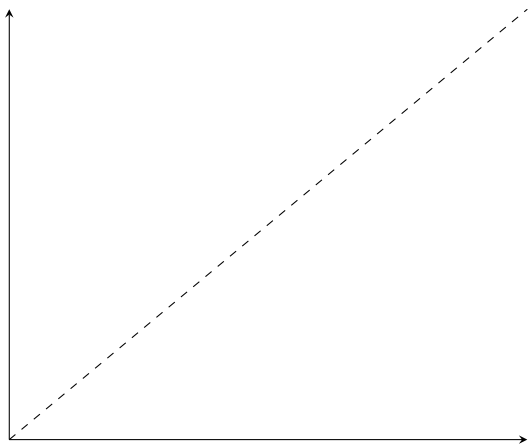
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

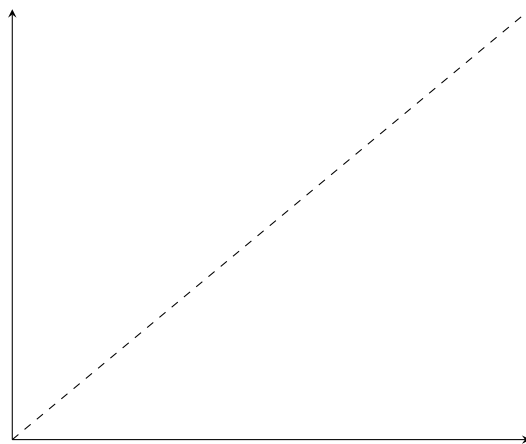
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$f_{2,1,2}^H$	4.0495788013554295	$t_{1,1,2}^H$	427.7238089674605
$z_{1,1,2}$	1.0	$f_{2,2,2}^H$	15.95042119864457	$t_{2,1,2}^H$	370.0
$z_{2,1,2}$	1.0	$f_{1,1,1}^C$	8.052417808908414	$t_{2,2,2}^H$	466.68393514135914
$z_{2,2,2}$	1.0	$f_{1,1,2}^C$	9.759368610010618	$t_{1,1,1}^C$	557.3655707854455
$z_{cu,1}$	1.0	$f_{2,1,2}^C$	5.240631389989382	$t_{1,1,2}^C$	520.6061775801204
$z_{cu,2}$	1.0	$f_{2,2,2}^C$	13.0	$t_{2,1,2}^C$	469.39382241987977
$z_{hu,1}$	1.0	$RecLMTD_{1,1,1}$	0.021166404292072658	$t_{2,2,2}^C$	500.0
$A_{1,1,1}^\beta$	18.13353447520507	$RecLMTD_{1,1,2}$	0.04321557847449181b	$t_{1,1,1}^{H,in}$	1308.5178939476182
$A_{1,1,2}^\beta$	13.825772197515043	$RecLMTD_{2,1,2}$	0.06700343198895575b	$t_{1,1,2}^{H,in}$	5936.33075337641
$A_{2,1,2}^\beta$	9.898970403313276	$RecLMTD_{2,2,2}$	0.009985451738077111b	$t_{2,1,2}^{H,in}$	2389.2514927997036
$A_{2,2,2}^\beta$	38.943261778500734	$RecLMTD_{cu,1}$	0.010753683011062897b	$t_{2,2,2}^{H,in}$	9410.748507200296
$A_{cu,1}^\beta$	9.28003023700696	$RecLMTD_{cu,2}$	0.010788088201949427b	$t_{1,1,1}^{H,out}$	744.8486473240288
$A_{cu,2}^\beta$	21.04660840930274	$RecLMTD_{hu,1}$	0.022585114716274293b	$t_{1,1,2}^{H,out}$	4277.238089674605
$A_{hu,1}^\beta$	12.595096067947068	$q_{1,1,1}$	563.6692466235894b	$t_{2,1,2}^{H,out}$	1498.344156501509
$A_{1,1,1}$	18.13353447520507	$q_{1,1,2}$	1659.0926637018051b	$t_{2,2,2}^{H,out}$	7460.7485072002955
$A_{1,1,2}$	13.825772197515043	$q_{2,1,2}$	890.9073362981949b	$t_{1,1,1}^{C,in}$	4670.40232916688
$A_{2,1,2}$	9.898970403313276	$q_{2,2,2}$	1950.0	$t_{1,1,2}^{C,in}$	4001.3411301043534
$A_{2,2,2}$	38.943261778500734	$q_{cu,1}$	577.2380896746054b	$t_{2,1,2}^{C,in}$	2148.6588698956466
$A_{cu,1}$	9.28003023700696	$q_{cu,2}$	1559.0926637018051b	$t_{2,2,2}^{C,in}$	4550.0
$A_{cu,2}$	21.04660840930274	$q_{hu,1}$	486.3307533764106b	$t_{1,1,1}^{C,out}$	5234.071575790469
$A_{hu,1}$	12.595096067947068	$t_{1,1}^{(H)}$	650.0	$t_{1,1,2}^{C,out}$	5660.433793806159
$dt_{1,1,1}$	32.42205022509404	$t_{1,2}^{(H)}$	593.6330753376411b	$t_{2,1,2}^{C,out}$	3039.566206193841
$dt_{1,1,2}$	13.63307533764106	$t_{1,3}^{(H)}$	427.7238089674605b	$t_{2,2,2}^{C,out}$	6500.0
$dt_{1,1,3}$	17.723808967460542	$t_{2,1}^{(H)}$	590.0		
$dt_{2,1,2}$	10.0	$t_{2,2}^{(H)}$	590.0		
$dt_{2,1,3}$	37.95463318509026	$t_{2,3}^{(H)}$	447.95463318509024		
$dt_{2,2,2}$	90.0	$t_{1,1}^{(C)}$	617.577949774906		
$dt_{2,2,3}$	97.95463318509026	$t_{1,2}^{(C)}$	580.0		
$dt_{cu,1}$	107.72380896746054	$t_{1,3}^{(C)}$	410.0		
$dt_{cu,2}$	127.95463318509026	$t_{2,1}^{(C)}$	500.0		
$dt_{hu,1}$	62.42205022509404	$t_{2,2}^{(C)}$	500.0		
$f_{1,1,1}^H$	2.013104452227105	$t_{2,3}^{(C)}$	350.0		
$f_{1,1,2}^H$	10.0	$t_{1,1,1}^H$	370.0		

3 *RecLMTD* Tangent Points

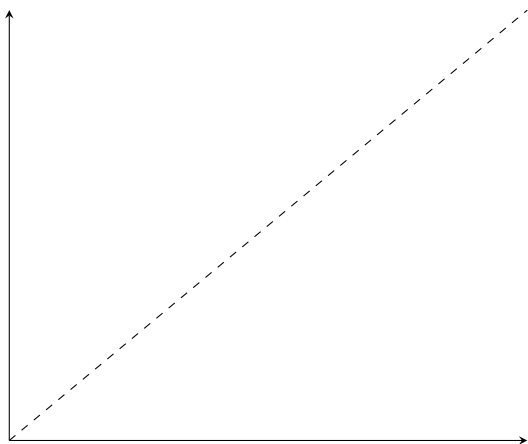
- $(1, 1, 1)$
 - Value: 0.021166
 - Error: 0.024942545701250
 - Relative error: 0.540948030802308
- $(1, 1, 2)$
 - Value: 0.043216
 - Error: 0.020931845564508
 - Relative error: 0.326308435265960
- $(2, 1, 2)$
 - Value: 0.067003
 - Error: 0.019290178835962
 - Relative error: -0.404293934310206
- $(2, 2, 2)$
 - Value: 0.009985
 - Error: 0.000661773910576
 - Relative error: 0.062154586782840
- $cu, 1$
 - Value: 0.010754
 - Error: 0.000673460667131
 - Relative error: 0.058935171036325
- $cu, 2$
 - Value: 0.010788
 - Error: 0.000380283033416
 - Relative error: -0.036538254440593
- $hu, 1$
 - Value: 0.022585
 - Error: 0.000014356791480
 - Relative error: 0.000635271115758



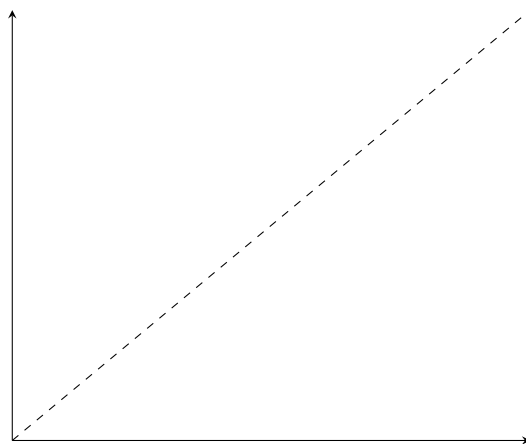
(a) *RecLMTD* - (1, 1, 1)



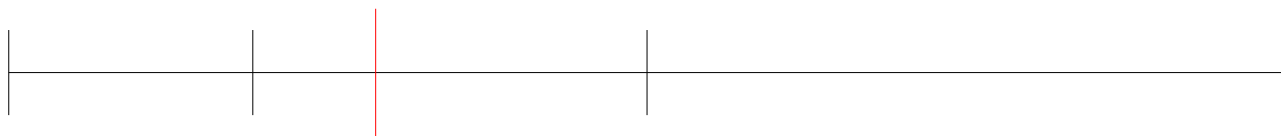
(b) *RecLMTD* - (1, 1, 2)



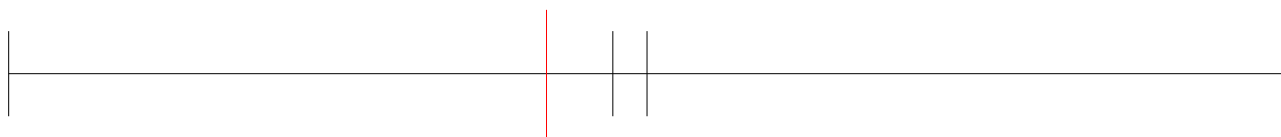
(c) *RecLMTD* - (2, 1, 2)



(d) *RecLMTD* - (2, 2, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1308.5178939476182
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 1, 2)
 - Value: 5936.33075337641
 - Absolute error: 9.094947017729282e-13
 - Relative error: -1.5320822567974912e-16
- (2, 1, 2)
 - Value: 2389.2514927997036
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 9410.748507200296
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 744.8486473240288
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 1, 2)
 - Value: 4277.238089674605
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 1498.344156501509
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 7460.7485072002955
 - Absolute error: 16.9431750546928
 - Relative error: 0.0022761442969934705

4.3 $b^{C,in}$

- (1, 1, 1)
 - Value: 4670.40232916688
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 1, 2)
 - Value: 4001.3411301043534
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 2148.6588698956466
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 5234.071575790469
 - Absolute error: 745.9311275253449
 - Relative error: 0.16620048684387365
- (1, 1, 2)
 - Value: 5660.433793806159
 - Absolute error: 579.6462061531183
 - Relative error: 0.11408589636018877
- (2, 1, 2)
 - Value: 3039.566206193841
 - Absolute error: 579.6462061531174
 - Relative error: 0.23563620204865257
- (2, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0



(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 1, 2)$



(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(H)} - (2, 2, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (2, 3)$



(a) $t^{(C)} - (1, 1, 1)$



(a) $t^{(C)} - (1, 1, 2)$



(a) $t^{(C)} - (2, 1, 2)$



(a) $t^{(C)} - (2, 2, 2)$

5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
 - Value: 18.133534
 - Error: 5.728167846880737
 - Relative error: 0.240056965323001
- $(1, 1, 2)$
 - Value: 13.825772
 - Error: 129.571526211802961
 - Relative error: 0.903584151508557
- $(2, 1, 2)$
 - Value: 9.898970
 - Error: 109.488727828922393
 - Relative error: 0.917085507553236
- $(2, 2, 2)$
 - Value: 38.943262
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000

5.2 A_{CU}

- $cu, 1$
 - Value: 9.280030
 - Error: 3.134840639537449
 - Relative error: 0.252506906492290
- $cu, 2$
 - Value: 21.046608
 - Error: 12.592649932751964
 - Relative error: 0.374343863491456

5.3 A_{HU}

- $hu, 1$
 - Value: 12.595096
 - Error: 0.585506958122934
 - Relative error: 0.044421864232225



(a) $A^\beta - (1, 1, 1)$



(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - (2, 2, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

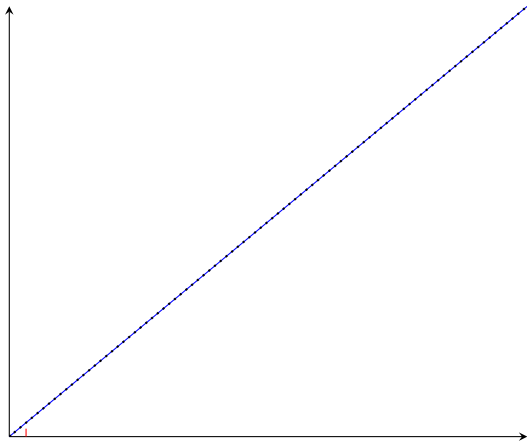


(a) $A^\beta - hu, 1$

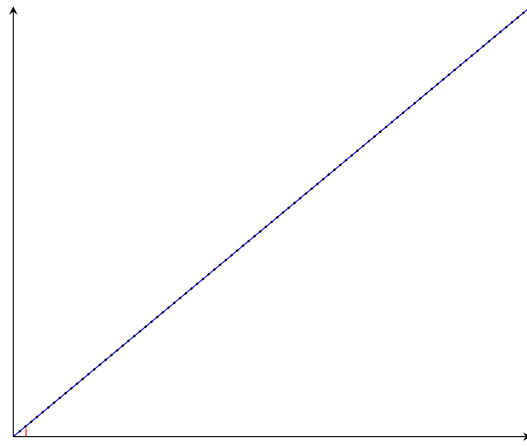
6 A^β Breakpoints

- (1,1,1)
 - q : 18.133534
 - $\hat{q}^\beta$: 18.133534
 - q^β : 18.133534
 - Breakpoints: [0, 5.0, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1,1,2)
 - q : 13.825772
 - $\hat{q}^\beta$: 13.825772
 - q^β : 13.825772
 - Breakpoints: [0, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,1,2)
 - q : 9.898970
 - $\hat{q}^\beta$: 9.898970
 - q^β : 9.898970
 - Breakpoints: [0, 28.33333333333332, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,2,2)
 - q : 38.943262
 - $\hat{q}^\beta$: 38.943262
 - q^β : 38.943262
 - Breakpoints: [0, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 9.280030
 - $\hat{q}^\beta$: 9.280030
 - q^β : 9.280030
 - Breakpoints: [0, 2.9819181007907165, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$

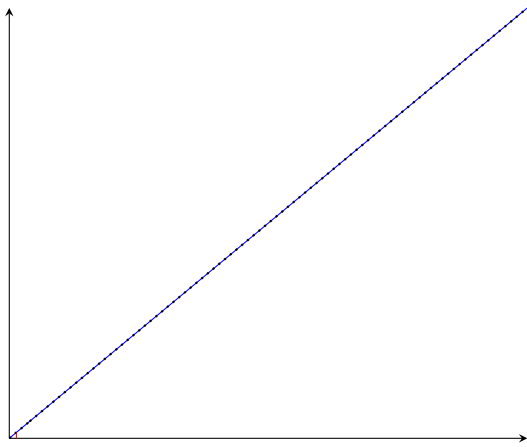
- q : 21.046608
- $\hat{q}^\beta$: 21.046608
- q^β : 21.046608
- Breakpoints: [0, 24.97364426355679, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 12.595096
 - $\hat{q}^\beta$: 12.595096
 - q^β : 12.595096
 - Breakpoints: [0, 4.943755299006499, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



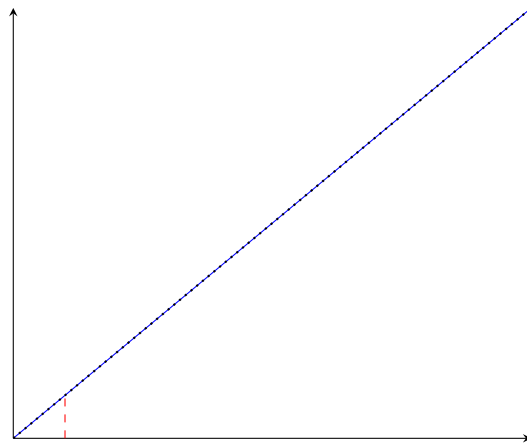
(a) $q^\beta - (1, 1, 1)$



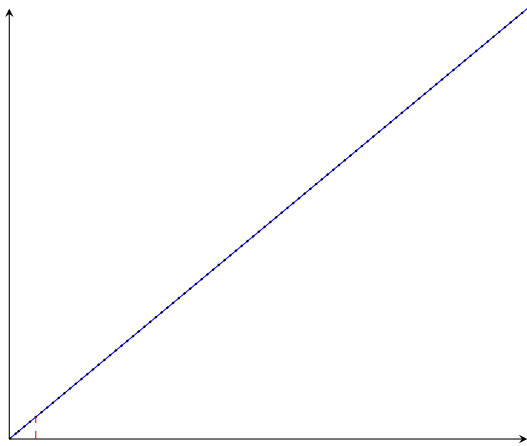
(b) $q^\beta - (1, 1, 2)$



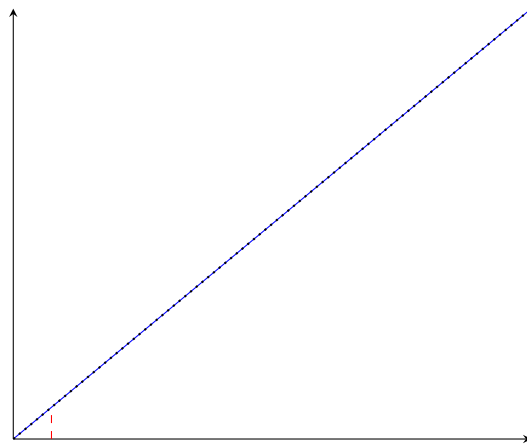
(c) $q^\beta - (2, 1, 2)$



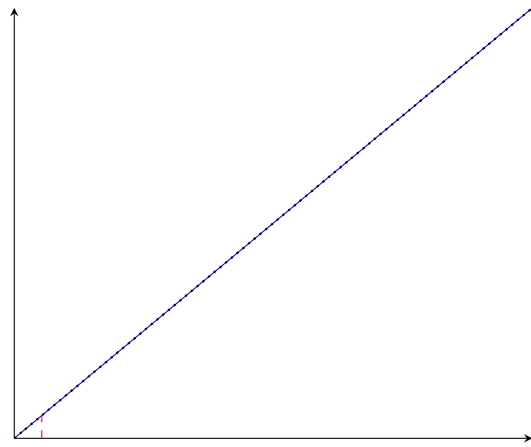
(d) $q^\beta - (2, 2, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value
$z_{1,2,1}$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255
$z_{1,2,2}$	-0.0	$q_{1,2,1}$	0.0
$z_{2,1,1}$	-0.0	$q_{1,2,2}$	0.0
$z_{2,2,1}$	-0.0	$q_{2,1,1}$	0.0
$z_{hu,2}$	-0.0	$q_{2,2,1}$	0.0
$A_{1,2,1}^\beta$	0.0	$q_{hu,2}$	0.0
$A_{1,2,2}^\beta$	0.0	$H_{1,2,1}$	650.0
$A_{2,1,1}^\beta$	0.0	$H_{1,2,2}$	370.0
$A_{2,2,1}^\beta$	0.0	$H_{2,1,1}$	590.0
$A_{hu,2}^\beta$	0.0	$H_{2,2,1}$	590.0
$A_{1,2,1}$	0.0	$C_{1,2,1}$	500.0
$A_{1,2,2}$	0.0	$C_{1,2,2}$	350.0
$A_{2,1,1}$	0.0	$C_{2,1,1}$	580.0
$A_{2,2,1}$	0.0	$C_{2,2,1}$	350.0
$A_{hu,2}$	0.0	$H_{1,2,1}^{in}$	5191.482106052382
$dt_{1,2,1}$	150.0	$H_{1,2,2}^{in}$	0.0
$dt_{1,2,2}$	223.63307533764112	$H_{2,1,1}^{in}$	11800.0
$dt_{1,2,3}$	207.72380896746054	$H_{2,2,1}^{in}$	0.0
$dt_{2,1,1}$	24.166666666666668	$H_{1,2,1}^{out}$	5191.482106052382
$dt_{2,2,1}$	90.0	$H_{1,2,2}^{out}$	0.0
$dt_{hu,2}$	180.0	$H_{2,1,1}^{out}$	11800.0
$f_{1,2,1}^H$	7.986895547772895	$H_{2,2,1}^{out}$	0.0
$f_{1,2,2}^H$	0.0	$C_{1,2,1}^{in}$	6500.0
$f_{2,1,1}^H$	20.0	$C_{1,2,2}^{in}$	0.0
$f_{2,2,1}^H$	0.0	$C_{2,1,1}^{in}$	4029.5976708331195
$f_{1,2,1}^C$	13.0	$C_{2,2,1}^{in}$	0.0
$f_{1,2,2}^C$	0.0	$C_{1,2,1}^{out}$	6500.0
$f_{2,1,1}^C$	6.947582191091586	$C_{1,2,2}^{out}$	0.0
$f_{2,2,1}^C$	0.0	$C_{2,1,1}^{out}$	4029.5976708331195
$RecLMTD_{1,2,1}$	0.005127303322837854	$C_{2,2,1}^{out}$	0.0
$RecLMTD_{1,2,2}$	0.004270239531638324		
$RecLMTD_{2,1,1}$	0.02916666666666668		
$RecLMTD_{2,2,1}$	0.010239999999999999		